

Alejandro Marchesini

Los Angeles, CA | alejandro.marchesini12@gmail.com | (562) 550-8904 | [LinkedIn](#) [Portfolio](#) [GitHub](#)

Skills

Programming Languages: Python, SQL, LaTeX

Data & ML: Data Cleaning, EDA, Feature Engineering, Machine Learning (XGBoost, Scikit-learn), Monte Carlo Simulation, Dashboard Design, API Data Ingestion, Statistical Analysis

Tools & Technologies: Git/GitHub, Jupyter Notebook, pandas, NumPy, Matplotlib, Tableau, Overleaf

Experience

Data Scientist

September 2025 - May 2026

Crowell+ Digital Marketing Group

La Mirada, CA

- Explored an automated YouTube data pipeline via Google APIs and BigQuery; diagnosed credential blockers and pivoted to manual extraction through YouTube Studio analytics.
- Collected and analyzed multi-year video performance and audience demographic data, consolidating findings into a structured Google Sheets model with charts and trend visualizations.
- Delivered a strategic playbook to the university marketing manager outlining 7+ data-driven hypotheses on audience behavior, content performance, and channel growth opportunities.

Math & Computer Science Intern

June 2025 - Aug 2025

Biola University

La Mirada, CA

- Led prototyping of 4+ equation formats across LaTeX, MathJax, and Overleaf, building the foundational, maintainable code for an educational app that generates readable math expressions for children.
- Facilitated weekly meetings to present updates, incorporate feedback, and refactor code toward technical and pedagogical goals.
- Developed 6+ LaTeX documents (100-200 lines) over 45 hours, building strong proficiency in LaTeX from scratch.

Projects

FIFA 2026 World Cup ML Simulation

February 2026 - May 2026

Python, XGBoost, Scikit-learn | [GitHub repo](#)

La Mirada, CA

- Built an end-to-end ML pipeline using pandas and NumPy to merge and clean ~47,000 historical matches with FIFA ranking data, applying time-aware joins to prevent data leakage.
- Engineered 7 features from 31 years of match data via EDA; applied a temporal train/test split across 3 classifiers (Logistic Regression, Random Forest, XGBoost).
- Achieved 59.18% accuracy with XGBoost; interpreted model behavior through feature importance analysis, identifying rank difference (37%) and FIFA points (22%) as top predictors — a lightweight explainability approach.
- Ran 10,000 Monte Carlo simulations of the 48-team bracket, with Argentina emerging as the favorite at 9.13%

Olympic Swimming Performance Analysis

April 2025 - May 2025

Excel & Tableau | [GitHub repo](#)

La Mirada, CA

- Integrated 2 datasets and web records into a unified Excel model, enabling consistent historical analysis.
- Cleaned & standardized 50+ years/13 Olympic cycles of performance and world-record data to ensure comparability.
- Developed an interactive Tableau dashboard visualizing performance trends by gender, stroke, and distance, with 100% of data integrated for comprehensive insights.
- Created a % Behind World Record metric to improve cross-event comparisons and reveal progression trends.

Education

Bachelor of Computer Science: Data Science and Information Systems | GPA: 3.97

May 2026

Biola University, La Mirada, CA

- **President's Scholarship | Elite 18 Award** – highest cumulative GPA on the roster (2024-2025)
- **Relevant Coursework:** Software Engineer, Data Science, Database Management, Business Analytics, Data Structures, UI/UX, Operating Systems, Computer Communications, Business Finance.
- **NCAA Division II Student-Athlete - Swimming - Team Culture Leader:** Committed 20+ hours per week to training, competition, and travel while fostering accountability and positive team dynamics.